BHARAT MISHRA

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EDUCATION

- **Bachelor of Technology, VIT Bhopal University** | Computer Science Engineering | **CGPA: 9.05** [2023 - 2027]
- **Grade XII, Vidyatree Modern World School** | Percentage: 95.50 [2021 - 2023]

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Java, SQL
- **Backend and Systems:** FastAPI, REST APIs, Docker
- **AI / ML:** TensorFlow, Scikit-learn, OpenAI Gym
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Neural Networks, Reinforcement Learning, Embeddings, NLP & LLM Pipeline

PROJECTS

DepoIndex: AI-Powered Legal Document Indexing | Python, LLM APIs, NLP, Prompt Engineering, PDF Processing

- Built a document processing pipeline to extract structured topic indices from legal deposition PDFs.
- Designed preprocessing logic to annotate text with page and line metadata, enabling accurate source referencing.
- Optimized LLM input formatting to reduce token usage while preserving extraction accuracy.
- Achieved 95% page-level and 85% line-level accuracy in topic referencing through manual validation.

Attentia.ai- Adaptive Attention Aware learning system | Python, FastAPI, OpenCV, MediaPipe, Reinforcement Learning

- Designed a real-time decision system to process user attention signals and trigger adaptive interventions with low-latency response.
- Implemented multi-modal signal processing using OpenCV and MediaPipe, combining facial landmarks, head pose, and audio inputs.
- Built a personalized baseline calibration mechanism to detect relative deviations in user engagement during live sessions.
- Defined backend data flow and API interaction patterns to efficiently process streaming inputs.
- Developed FastAPI endpoints to integrate real-time inputs with the decision engine.
- Modeled intervention logic using tabular Q-learning to adapt system responses under stochastic user behavior.

MovieMix: Latent Semantic Movie Search Engine (MovieLens 25M) | Python, FastAPI, Docker, NumPy, Embeddings, Recommendation Systems

- Built a semantic search system that maps user input tags to vector embeddings and retrieves relevant movies using cosine similarity.
- Designed a custom embedding pipeline to address cold-start scenarios using tag-based representations.
- Developed and deployed a FastAPI-based inference service with Docker, achieving sub-50ms response time.

Search and Rescue Reinforcement Learning System | Python, Reinforcement Learning, Custom Environment

- Simulated a grid-based environment and implemented agents using Q-learning and PPO for path optimization.
- Defined state space, action space, and reward functions to model multi-agent navigation.
- Reduced target search time by ~90% compared to a random baseline.

ACHIEVEMENTS & CERTIFICATIONS

- **Global Rank 8261, TCS CodeVita (2024):** Advanced to the penultimate round by solving complex algorithmic problems.
- **Generative AI Training (Kaggle/Google) (2024):** Transformers, LLMs, and prompt engineering.
- **Google Certified: Computer Networking:** The Bits and Bytes of Computer Networking.

LEADERSHIP AND VOLUNTEERING

- **PR and Outreach Lead, Startup Club:** Secured industry partnerships for guest lectures and sponsorships by coordinating directly with C-suite executives and other industry professionals.
- **Volunteer, Pahal Foundation:** Taught basic mathematics and fundamental elementary education to underprivileged children.